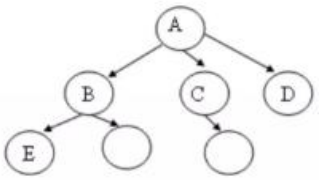
Tree

Introduction to trees

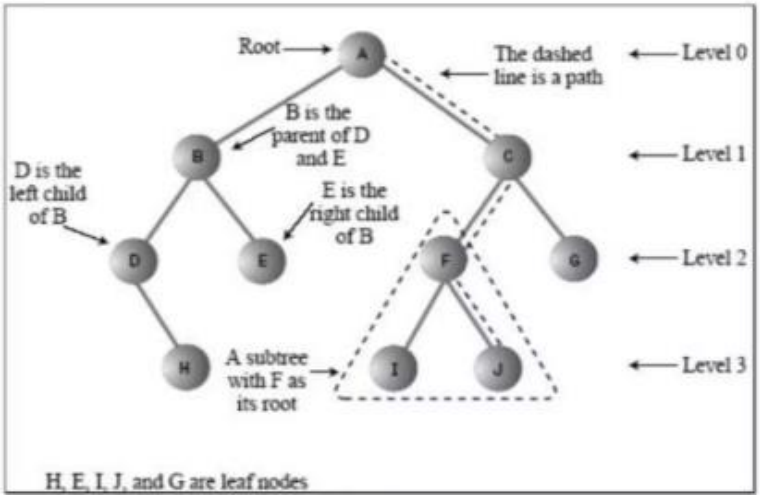
- So far we have discussed mainly linear data structures – strings, arrays, lists, stacks and queues
- Now we will discuss a non-linear data structure called **tree**.
- Trees are mainly used to represent data containing a hierarchical relationship between elements, for example, records, family trees and table of contents.
- Consider a parent-child relationship

Tree

- A tree is an abstract model of a hierarchical structure that consists of nodes with a parent-child relationship.
  - Tree is a sequence of nodes
  - There is a starting node known as a root node
  - Every node other than the root has a parent node.
  - Nodes may have any number of children



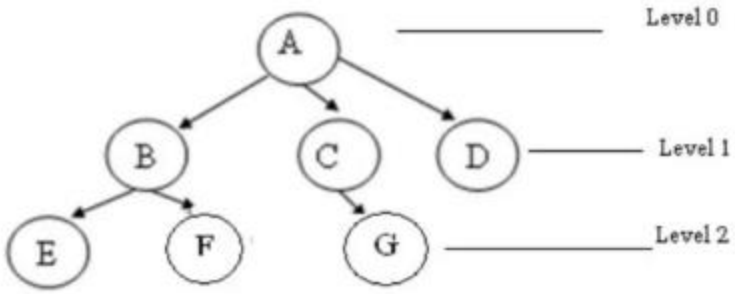
A has 3 children, B, C, D
A is parent of B



Some Key Terms:

- **Root** – Node at the top of the tree is called root.
- **Parent** – Any node except root node has one edge upward to a node called parent.
- **Child** – Node below a given node connected by its edge downward is called its child node.
- **Sibling** – Child of same node are called siblings
- **Leaf** – Node which does not have any child node is called leaf node.
- **Sub tree** – Sub tree represents descendants of a node.
- **Levels** – Level of a node represents the generation of a node. If root node is at level 0, then its next child node is at level 1, its grandchild is at level 2 and so on.
- **keys** – Key represents a value of a node based on which a search operation is to be carried out for a node.

- Degree of a node:
  - The degree of a node is the number of children of that node
- Degree of a Tree:
  - The degree of a tree is the maximum degree of nodes in a given tree
- Path:
  - It is the sequence of consecutive edges from source node to destination node.
- Height of a node:
  - The height of a node is the max path length form that node to a leaf node.
- Height of a tree:
  - The height of a tree is the height of the root
- Depth of a tree:
  - Depth of a tree is the max level of any leaf in the tree



- ✔ A is the root node
- ✔ B is the parent of E and F
- ✔ D is the sibling of B and C
- ✔ E and F are children of B
- ✔ E, F, G, D are external nodes or leaves
- ✔ A, B, C are internal nodes
- ✔ Depth of F is 2
- ✔ the height of tree is 2
- ✔ the degree of node A is 3
- ✔ The degree of tree is 3

Characteristics of trees

- Non-linear data structure
- Combines advantages of an ordered array
- Searching as fast as in ordered array
- Insertion and deletion as fast as in linked list
- Simple and fast